

WEEK 1: ORGAN SYSTEM

ANATOMY

- Structure of body
- Means to dissect, or cut apart and separate the parts of the body for study
- **Studies:** *body parts, microscopic organization, processes by how it develops, relationship between body parts and its function*

TWO BASIC APPROACH

- 1.) **Systemic** - cardiovascular, nervous, muscular, integumentary (introductory textbooks)
- 2.) **Regional** - organization by areas (e.g. head - skull, brain) (medical and dental schools)

ANATOMIST APPROACH

- 1.) **Surface** - external features (landmarks for locating deeper parts)
- 2.) **Anatomical Imaging** - use of x-rays, ultrasound, magnetic resonance imaging (MRI) (anything that creates pictures of internal structures)
- 3.) **Gross** - visible organs seen by the naked eye
- 4.) **Developmental** - changes occurred
- 5.) **Microscopy** - e.g. cytology, histology

PHYSIOLOGY

- Processes or function of living things
- "is dynamic, rather than fixed and unchanging"

GOALS:

- Understand and predict body's response to stimuli
- Understand how the body maintains homeostasis regardless of the changing internal and external environment

TYPES:

- Human physiology
- Cellular and systemic physiology - organizational level
- Pathophysiology - changes due to disease

IMPORTANCE OF KNOWING HUMAN ANATOMY AND PHYSIOLOGY

- 1.) Allows us to detect changes or stimuli, respond to stimuli, and perform many other actions.
- 2.) Understanding disease for treatments
- 3.) For student taking health sciences course to become a health care professional

STRUCTURAL AND FUNCTIONAL ORGANIZATION OF THE HUMAN BODY

1.) Chemical Level

- Structural and functional characteristic of an organism depend on their chemical makeup
 - Atoms -> Molecules

2.) Cell Level

- Basic structural and functional units of organisms
- Molecules -> Organelles (e.g. mitochondria) -> Cell
- Cells might be different in function and appearance, but they might be same in characteristic

3.) Tissue Level

- Combined by cells and materials surrounding it
- Types: **connective**, **muscle**, **epithelial**, **nervous** (CMEN)

4.) Organ Level

- Composed of two or more tissue type that performs one or more common functions

5.) Organ System Level

- Organs that belong in a group that works together
- 11 Kinds: integumentary, skeletal, muscular, lymphatic, respiratory, digestive, nervous, endocrine, cardiovascular, urinary, and reproductive
- Important for preventing consequences and maintaining homeostasis

6.) Organism Level

- Living thing considered as a whole

Atom -> Molecule -> Organelle -> Cell -> Tissue -> Organ -> Organ System -> Organism

CHARACTERISTIC OF A LIVING ORGANISM

1.) Movement

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- Different body parts working together to cause movement

2.) Metabolism

- Ability to use energy to perform functions (growth, movement, reproduction); sum of all chemical reactions

TWO TYPES:

- Catabolism - break down substances
- Anabolism - building up substances

3.) Responsiveness

- Ability to sense and react to change in the environment

- Includes movement towards food and away from danger

- E.g. when body experiences low blood pressure, it releases glucagon, which signals the liver to release the stored glucose to bring back the normal blood pressure

4.) Growth

- Increase in size (hypertrophy) of an organism due to increase of cell number (hyperplasia), size, and/or surrounding cells

- Mitosis: division producing two identical daughter cells

5.) Development

- Greatest development: before birth

- Goes with growth but with differentiation

- **Differentiation** - change in cell structure and function to generalized to specialized

- Makes up our different body parts

6.) Reproduction

- Formation of new cells or new organism

- Growth and tissue repair is impossible without it - extinction

HOMEOSTASIS

- Maintaining the balance in our body despite experiencing external and internal imbalance

- Cells are surrounded by small amount of fluid, and this fluid determines the function of a normal cell that might be affected by a **variable**

- *Variable are conditions that can change value*

- E.g. temperature, volume, chemical content

Homeostatic mechanisms

- **Shivering, sweating**

- Maintains ideal normal value or **set point**, and affects mostly by variables

- Governed by **nervous and endocrine**

- Body cannot maintain a solid, normal state every time because of unchanging environments. Our body then compromises in order to attain normal range

FEEDBACKS

1.) Negative feedback

- Compromises for the variables in order to maintain homeostatic balance within our body

- E.g. shivers and sweats - basically, homeostatic mechanism

- Components

- **Receptor (afferent)** - monitors the value of variable
- **Control center (efferent)** - establishes a set point, in which the variable is maintained
- **Effector** - changes the value of the variable
 - E.g. Sweating - thermoreceptor (receptor) detects change in temperature, which then sends a message to the hypothalamus (control center). The hypothalamus will then send a message that will trigger the blood vessels on the surface of the skin to relax, and signals sweat gland to produce sweat.
 - Thermoreceptor -> Hypothalamus -> Blood vessel on the surface of the skin relaxes -> Sweat gland produces sweat

2.) Positive feedback

- Tolerates stimulus to continue until it reaches it desired actions

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- E.g. giving birth - uterine contracts for the baby to get pushed, which then results for the uterus to stretch simultaneously until the baby is out

- The cervix will stretch first in the first contraction of labor, and will be detected by stretch-sensitive nerve cell, which then send an information to the brain to release oxytocin. The release of oxytocin will trigger more contractions from the uterus. Cycle continues until such time the fetus is out of the womb.
- E.g. blood clot - chemical responsible for clot will continue to reproduce until it forms clot (self-limiting)
- Detrimental e.g.
 - Loss of blood in an accident lowers the amount of blood, which affects the value of blood pressure within the body. Since the blood pressure drops, the heart cannot contract normally. If this cycle continues, it may cause death.
 - Donating blood is handled by negative feedback by speeding up the heartbeat, and restoring blood pressure. If negative feedback cannot keep up, positive feedback comes into play.

BODY POSITIONS

- 1.) **Anatomical position** - standing erect, face and palms facing front, limbs on the side
- 2.) **Fundamental position** - same as anatomical but the palms face towards the body
- 3.) **Supine/Prone**
- 4.) **Right lateral/left lateral (lateral recumbent)** - sidelying
- 5.) **Fowler's** - inclined
- 6.) **Trendelenburg** - declined

IMPORTANCE OF KNOWING BODY POSITION:

- The relativity of body parts depends on the position of the body
- E.g anatomical position - elbow is above the hand / prone and supine - elbow and hands are on the same level

DIRECTIONAL TERMS

- 1.) Superior/Cephalic - up
- 2.) Inferior/Caudal - down
- 3.) Anterior/Ventral - front
- 4.) Posterior/Dorsal - back
- 5.) Proximal - nearest (limbs)
- 6.) Distal - farthest (limbs)
- 7.) Medial - midline
- 8.) Lateral - away midline
- 9.) Superficial - surface
- 10.) Deep - interior

E.g., spleen is posterolateral to the stomach

E.g., optic chiasm is superior to the pituitary gland

BODY PARTS AND REGIONS

Cervical - 8; Thoracic - 12; Lumbar - 5; Sacral 5; Coccyx - 1

***Conus medullaris* - ends above L2**

1.) Head

- Frontal (forehead)
- Orbital (eyes)
- Nasal (nose)
- Oral (mouth)
- Otic (ear)
- Mental (chin)
- Buccal (cheek)
- Occipital (base of skull)
- Cranial (skull)
- Nuchal (back of neck)

2.) Neck - Cervical

3.) Trunk

A.) Thoracic (thorax)

- Pectoral (chest)
- Sternal (breastbone)
- Mammary (breast)

B.) Abdominal (abdomen)

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- **4 quadrant** - right and left upper, right and left lower
- **9 regions** - right and left hypochondriac, epigastric, right and left lumbar, umbilical, right and left iliac, hypogastric

- Pelvic (pelvis) - *L4 (iliac crest)*
- Umbilical (navel) - *T10*
- Pubic (genitals)
- Inguinal (groin)

C.) Back

- Scapular (shoulder blade)
- Vertebral (spinal column)
- Lumbar (loin)
- Sacral (between hips)
- Gluteal (buttock) - *S2*
- Perineal (perineum)

4.) Upper Limb

A.) Front

- Clavicular (collarbone)
- Axillary (armpit)
- Brachial (arms)
- Antecubital (front of elbow)
- Antebrachial (forearm)
- Carpal (wrist)
- Palmar (palm)
- Digital (fingers)

B.) Back

- Acromial (point of shoulder)
- Olecranon (point of elbow)
- Dorsum (back of hand)

5.) Lower Limb

A.) Front

- Coxal (hip)
- Femoral (thigh)
- Patellar (kneecap)
- Crural (leg)
- Talus/Tarsal (ankle)
- Dorsum (top of foot)
- Digital (fingers)
- Pedal (foot)

B.) Back

- Popliteal (back of patellar)

- Sural (calf/back of crural)
- Plantar (sole)
- Calcaneal (heel)

PLANES

- Used for looking inside and observing body structures

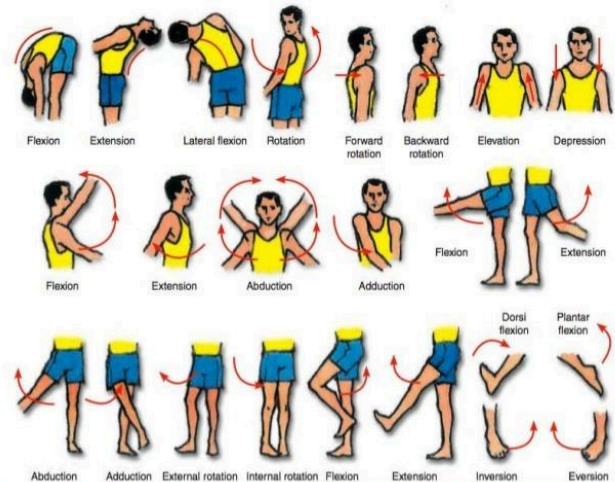
KINDS:

- 1.) Sagittal plane - mid, y-axis, flight of an arrow, left and right, flexion-extension
 - Median plane - directly middle
 - Parasagittal - 2 cm cut lateral to median plane
- 3.) Transverse plane - **horizontal**, x-axis, up and down, rotation
- 4.) Frontal plane - **coronal**, z-axis, front and back, abduction-adduction

IF IN ORGAN:

- 1.) Longitudinal section - up and down, through long axis
- 2.) Transverse section - left and right, right angle along long axis
- 3.) Oblique section - diagonal, cut other than right angle along long axis

BODY MOVEMENTS



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BODY CAVITIES

1.) Thoracic cavity

- Surrounded by rib cage
- Diaphragm - separates abdominopelvic cavity and thoracic cavity
- **Mediastinum** - divides thoracic cavity into left and right parts
 - Blood vessel
 - Heart
 - Esophagus
 - Lungs
 - Trachea
 - Thymus

2.) Abdominal cavity

- Bounded by abdominal muscle
- Contains the stomach, spleen, liver, intestines, pancreas, kidney

3.) Pelvic cavity

- Enclosed by pelvic bones
- Contains urinary bladder, part of large intestine, internal reproductive organ

Serous Membrane

- Lines the trunk cavity and covers the organ within

PARTS:

1.) **Visceral serous membrane** - covers *internal organ*

2.) **Parietal serous membrane** - covers the *walls of the cavity*

3.) **Serous fluid** - space between the membrane, *thin, lubricating film of serous fluid* produced by the membranes

A.) Thoracic cavity (heart)

- **Pericardial cavity** - surrounds the heart, contains pericardial fluid
- **Parietal pericardium** - covers the heart, connective tissue sac
- **Visceral pericardium** - lines the inside of the heart

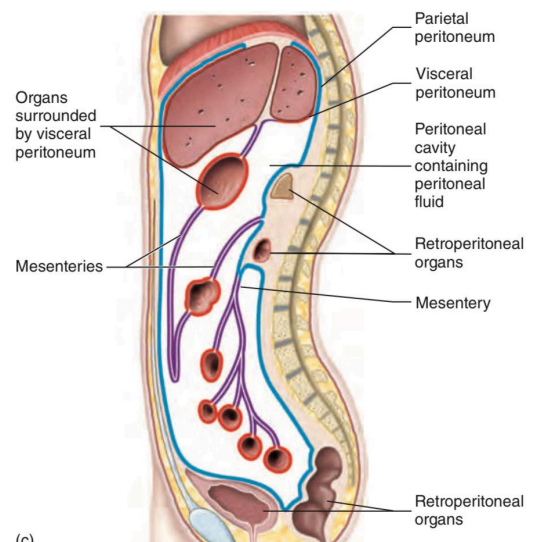
B.) Pleural cavity (lungs)

- **Parietal pleura** - lines the inner surface of thoracic wall, lateral of mediastinum

- **Visceral pleura** - lines the lung wall
- **Pleural cavity** - contains pleural fluid

C.) Abdominopelvic cavity

- **Peritoneal cavity** - contains peritoneal fluid
- **Visceral peritoneum** - covers the organ within abdominopelvic cavity
- **Parietal peritoneum** - lines the abdominopelvic cavity, inferior to diaphragm
- **Mesenteries**
 - Two peritoneum fused together the runs through visceral peritoneum of some organ up to the parietal peritoneum on the body wall
 - Provides pathway for nerves and blood vessel for the abdominopelvic organ
 - Anchors organ to the body wall
- **Retroperitoneal organ** - found behind
 - Abdominopelvic organ is closely attached to the body wall and do not have mesenteries
 - Covered by parietal peritoneum
 - Pancreas, part of intestine, adrenal gland, kidney, urinary bladder



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DISEASES ASSOCIATED:

Pericarditis - pericardium

Pleurisy - pleura

Peritonitis - peritoneum

ORGAN SYSTEMS

1.) Integumentary System

- Protection, regulates temperature, prevents water loss, helps produce vitamin D
- Hair, nails, skin, sweat glands

2.) Skeletal System

- Support and protection, allows body movements, produces blood cells, store minerals and adipose tissue
- Bones, ligaments, joints, cartilages

3.) Muscular System

- Regulates temperature, produce movements, maintain postures
- Tendons - connection between muscle and bones

4.) Lymphatic System

- Removes foreign substances from the blood and lymph, fights diseases, maintains tissue fluid balance, absorbs dietary fats
- Lymph nodes, lymphatic vessels, other lymphatic organs

5.) Respiratory System

- O₂-CO₂ exchange, regulates blood pH
- Lungs, respiratory passages

6.) Digestive System

- Mechanical and chemical digestion, absorbs nutrients, removes waste
- Mouth, esophagus, stomach, intestine, accessory organs

7.) Nervous System

- Detects sensation and controls movements, physiological processes, and intellectual functions
- Brain, spinal cord, nerves, sensory receptors

8.) Endocrine System

- Metabolism, growth, reproduction
- Glands, hormones

9.) Urinary System

- Removes waste from the blood, regulates blood pH, ion balance, and water balance.
- Kidneys, urinary bladder, ducts

10.) Female Reproductive System

- Produces oocyte, site of fertilization, produces milk, produces hormone (estrogen, progesterone)
- Uterus, vagina, uterine tubes, mammary glands

11.) Male Reproductive System

- Produces and transfers sperm cells, produces hormone (testosterone)
- Penis, accessory structures, ducts, testes

12.) Cardiovascular

- Transport nutrients, waste products, gases, and hormones, play a role in the immune response and regulation of body temperature
- Heart, blood, blood vessels.

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